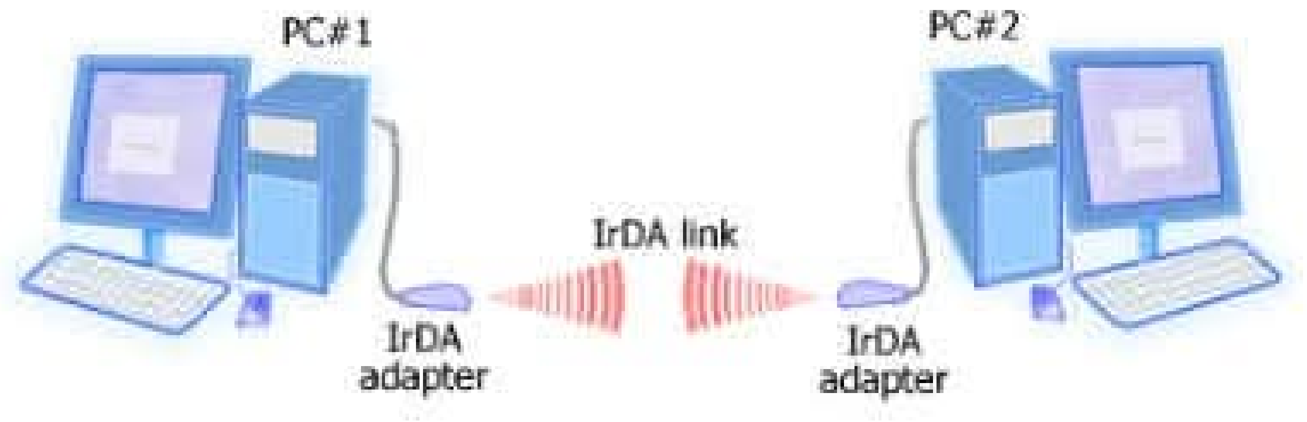
# IrDA Communication

Infrared Data Association - Light-Based Communication

# What is IrDA?

## Think of IrDA like a TV remote that can send files!



**Real-world analogy:**

- Like using a flashlight to send Morse code messages
- Must have clear line-of-sight (no obstacles)
- Works only in the dark (invisible infrared light)
- Fast blinking = data transmission

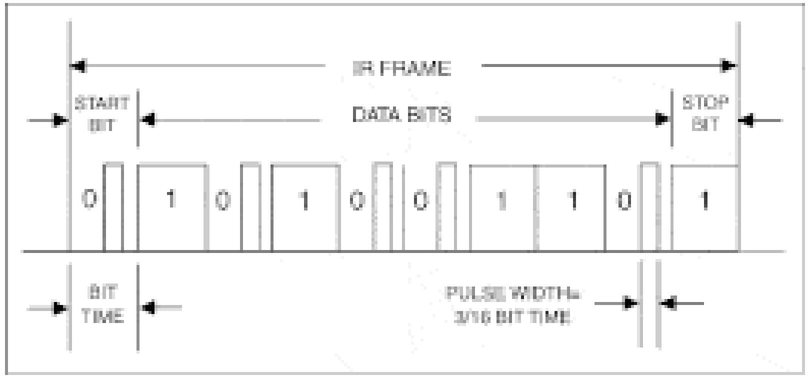
# How IR Works: The Flashlight Analogy

## Basic Principle:

```
Device A                        Device B
               IR Light
 ┌──────┐                       ┌──────┐
 │ LED  │  ))))))))))))))       │ Photo│
 │(Send)│                       │(Recv)│
 └──────┘                       └──────┘
```

## Like Morse Code with invisible light:

- **LED ON** = Binary '1'
- **LED OFF** = Binary '0'




## Example:

```
Data: "Hi" = 01001000 01101001
LED:  OFF–ON–OFF–OFF–ON–OFF–OFF–OFF OFF–ON–ON–OFF–ON–OFF–OFF–ON
```

Simple and effective way to transmit data with security (line-of-sight) and low power!

## Physical Layer: Invisible Light

IR LED Wavelength:

```
┌─────────────────────────────────────┐
│ Visible Light │  Infrared (IR)      │
│ 400–700 nm    │  700–1000 nm        │
└─────────────────────────────────────┘
                 │
                 ├─ 850 nm (common)
                 ├─ 940 nm (TV remotes)
                 └─ 950 nm (data)
```

Human eye: Cannot see IR (completely invisible)
IR receiver: Detects IR light as electrical signal
Camera: Can see IR (try pointing remote at phone camera!)

LED Output Power: 20-100 mW
Typical range: 1-10 meters (depends on LED power)

**Test:** Point TV remote at phone camera while pressing button - you'll see the LED flash!

## Why Infrared?

Pros:

- **Invisible** - doesn't disturb users
- **Low power** - efficient for battery devices
- **Abundant** - cheap LEDs and photodiodes
- **Secure** - can't penetrate walls

Cons:

- **Line-of-sight required** - must point devices at each other
- **Short range** - typically 1-2 meters

# IR Modulation: Carrier Frequency

Sunlight contains IR! How to filter it out?

Solution: Modulate IR at specific frequency (38 kHz typical)

```
Without modulation (DC):
IR     ────────────────   ← Can't distinguish from sunlight
       ON all the time

With 38 kHz modulation:
IR     ▄▀▄▀▄▀▄▀▄▀▄▀▄▀▄▀   ← Receiver only responds to 38 kHz
       38,000 pulses/sec
```

Receiver has built-in bandpass filter:

- Only passes 36-40 kHz signals
- Blocks DC sunlight
- Blocks 50/60 Hz fluorescent flicker

# Before IrDA (TV Remote Era)

## IR Standards & Frequencies

- 36 kHz: Philips RC-5/RC-6
- 38 kHz: (Most common worldwide) NEC protocol & Default for most devices
- 40 kHz: Some Sony devices (SIRC)
- 56 kHz: RC-6 (Microsoft MCE)

# Most Common Encoding: Pulse Distance

NEC Protocol Example:

```
Binary 0:
   ▄▀_
  562µs ON, 562µs OFF = 1.125 ms total

Binary 1:
   ▄▀____
  562µs ON, 1.687ms OFF = 2.25 ms total                                          .

 ┌──────────────────────────────────────────────────────┐
 │ Start   │ Address  │ ~Address│ Command │~Command│
 │ 9ms     │ 8 bits   │ 8 bits  │ 8 bits  │ 8 bits │
 └──────────────────────────────────────────────────────┘
```

Longer space = 1, Shorter space = 0

**Example: Power button on Samsung TV**

- Address: 0x04
- Command: 0x12
- Packet: 0x04FB12ED (includes inverses)

**Timing:**

- Start: 9 ms burst + 4.5 ms space
- Bit 0: 562 µs burst + 562 µs space
- Bit 1: 562 µs burst + 1.687 ms space
- Total time: ~67 ms per button press

# IrDA History: Before Bluetooth and WiFi

## The Problem (1990s):

```
Laptop ---- No easy way to share files ---- Another Laptop
PDA --------- Difficult data sync --------- Computer
Phone ------- No wireless transfer ------- Computer
```

## IrDA Solution (1993-2010):

```
Laptop )))IR LIGHT((( Another Laptop
PDA    )))IR LIGHT((( Computer
Phone  )))IR LIGHT((( Computer
```

## IrDA was the first mainstream wireless data communication!

# IrDA Speed Evolution

| Mode | Full Name | Speed | Relative Speed | 3MB Transfer Time | Experience |
|---|---|---|---|---|---|
| SIR | Serial IR | 115.2 Kbps | 1× | ~4 minutes | Very slow |
| MIR | Medium IR | 1.152 Mbps | 10× | ~25 seconds | Acceptable |
| FIR | Fast IR | 4 Mbps | 35× | ~6 seconds | Fast |
| VFIR | Very Fast IR | 16 Mbps | 140× | ~1.5 seconds | Very fast |

# IrDA vs TV Remote: What's the Difference?

## TV Remote (Simple):

```
Remote ))))))) TV
- One-way communication
- Simple commands (ON/OFF/CHANNEL)
- Custom protocols per manufacturer
- ~38kHz carrier frequency
```

## IrDA (Advanced):

```
Device A ))))))) Device B
        ((((((
- Two-way communication
- Complex data (files, documents)
- Standardized protocol
- Multiple speed modes
```

**IrDA is like upgrading from smoke signals to telephone!**

# IrDA Protocol Stack

```
┌──────────────────────┐
│     Applications     │  ← File transfer, Print, etc.
├──────────────────────┤
│        IrCOMM        │  ← Serial/Parallel port emulation
├──────────────────────┤
│        IrLMP         │  ← Link Management (like TCP)
├──────────────────────┤
│        IrLAP         │  ← Link Access Protocol (like MAC)
├──────────────────────┤
│    Physical (IR)     │  ← Infrared light transmission
└──────────────────────┘
```

**Similar to network stack, but much simpler!**

## Physical Layer: The Light Spectrum

### Infrared Light Characteristics:

```
Visible Light Spectrum:
Red —— Orange —— Yellow —— Green —— Blue —— Purple
700nm                                          400nm

Infrared (IrDA):
850nm ——— 900nm (Invisible to human eyes)
```

## Connection Establishment:

```
Device A            Device B
   |                  |
   |--- Discovery -->|  (Who's there?)
   |<-- Response ----|  (I'm here!)
   |--- Connect  --->|  (Let's talk!)
   |<-- Accept   ----|  (OK!)
   |=== Data ========|  (File transfer)
   |--- Disconnect ->|  (Goodbye!)
```

## Error Handling:

- **Checksums** for data integrity
- **Retransmission** for failed packets
- **Flow control** to prevent overflow

# Limitations: Why IrDA Faded Away

**Physical Limitations:**

- Line-of-sight required: Can't work through walls, around corners
- Distance limited: Typically 1-2 meters maximum
- Sensitive to interference: Sunlight, fluorescent lights cause problems
- Alignment critical: Devices must be precisely positioned

**Practical Problems:**

- Users forgot to align devices properly
- Sunlight interference outdoors
- Slow transfer speeds compared to emerging technologies
- Both devices needed IrDA ports

# Modern Infrared beyond IrDA

## TV/Entertainment Remotes

TV Remote )))IR((( Smart TV

- Simple, reliable, cheap
- No pairing required
- Works through coffee tables

# TV Remote Evolution

| Feature | Traditional IR Remote | Smart TV Remote (Modern) |
| --- | --- | --- |
| Primary Connection | Infrared (IR) | Bluetooth (often) |
| Backup Connection | None | IR (usually for power) |
| Line-of-Sight Required | Yes | No (Bluetooth) |
| Pairing Needed | No | Yes |
| Communication | One-way | Two-way |
| Voice Control | No | Yes (mic + assistant) |
| Pointer / Motion | No | Yes (gyro / air mouse) |
| Battery Reporting | No | Yes |
| Reliability | Very simple & stable | More features, more complexity |

## Industrial Applications Using IR

- Temperature Gun
- Night Vision
- Military, surveillance
- Infrared illumination
- LiDAR (Light Detection and Ranging)

# IrDA vs Modern Alternatives

| Feature | IrDA | Bluetooth | WiFi | USB |
|---|---|---|---|---|
| **Setup** | Point & align | Pair once | Network setup | Plug cable |
| **Range** | 1-2 meters | 10 meters | 100+ meters | Cable length |
| **Speed** | Up to 16 Mbps | Up to 3 Mbps | 100+ Mbps | 480 Mbps+ |
| **Power** | Very low | Low | Medium | No battery needed |
| **Security** | Line-of-sight | Encryption | WPA2/WPA3 | Physical access |
| **Obstacles** | None allowed | Some OK | Most OK | None |

# IrDA on Arduino — Practical Reality

## Full IrDA Stack on Arduino?

❌ Not practical

- IrDA requires:
  - Strict pulse timing (SIR 3/16 bit encoding)
  - Link management (IrLAP)
  - Optional upper layers (IrLMP, IrCOMM)

- AVR-based Arduino:
    - Limited RAM
    - Limited timing precision
    - No native IrDA hardware support
- MIR / FIR speeds → unrealistic

👉 Implementing the full IrDA protocol stack in software is overly complex.

## Simple IR Communication?

✅ Very reasonable

- IR LED + IR receiver
- Custom protocol (UART over IR, Manchester, etc.)
- Low cost
- Easy to prototype
- Good for short-range device-to-device communication

👉 If you just need infrared communication, skip IrDA.

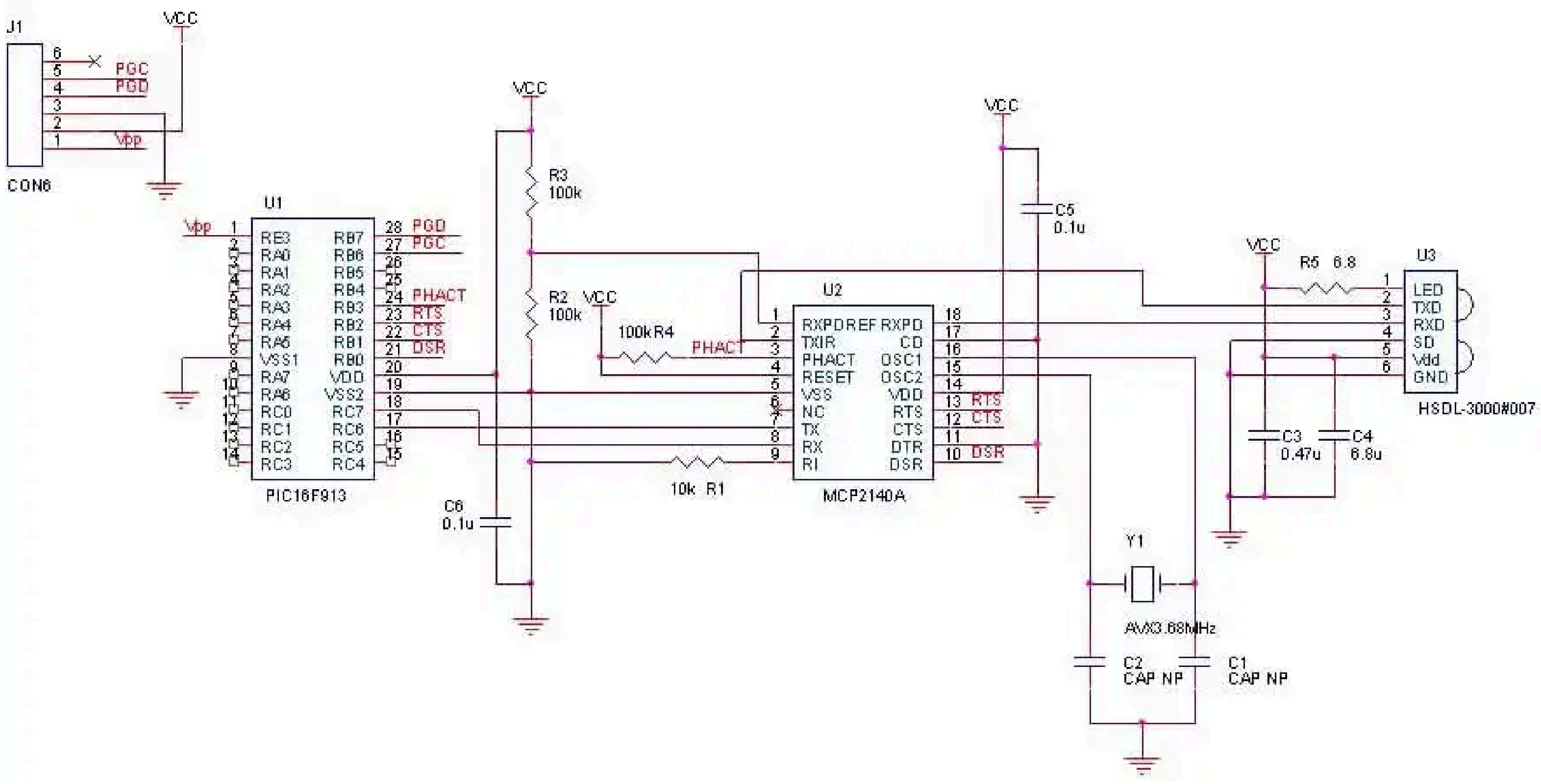
## Want IrDA-Style Serial?

Use an external controller like:

**MCP2140**

- Handles IrDA protocol
- Connects via UART
- Appears like a serial cable replacement (IrCOMM style)
- Offloads protocol complexity from Arduino

# Arduino = normal Serial
# MCP2140 = IrDA translator



## Bottom Line

- Full IrDA stack on Arduino → ❌ Not practical
- Simple IR link → ✅ Easy and reasonable
- True IrDA behavior → Use MCP2140 as a serial bridge

# IR (not IrDA) and Modern IoT Applications

- Sensor data transfer (short-range)
- Remote controls (TVs, ACs)
- Point-to-point communication (e.g. cash registers)
- Secure communication (line-of-sight)
- Line-following robots (using IR sensors)
- Proximity detection (e.g. automatic doors)